NUCLEO-H743ZI2 VDD_MCU PDN Review

3.3V Rail · STM32H743ZIT6U · MB1364 Rev C01

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This report covers

- Target impedance derivation from droop budget and STM32H743 datasheet current spec
- Decoupling BOM extracted from ST's published MB1364 schematic (fully traceable)
- PDN impedance profile: bare vs decoupled, DC to 500MHz
- Anti-resonance analysis at each capacitor tier transition
- Verdict with margin analysis and production VRM candidate recommendations

Sample deliverable. For a custom PDN review of your board, contact via Fiverr.

1. Scope and Design Parameters

This review analyses the **VDD_MCU (3.3V)** power delivery network of the NUCLEO-H743ZI2 development board (MB1364 Rev C01, STMicroelectronics). The board also carries a 1.8V rail and an internal 1.2V VCORE LDO. This review focuses on VDD_MCU as the primary external PDN rail with a full designer-controlled decoupling network. A multi-rail review is available as a separate engagement.

> **Rail selection note:** The STM32H743 VCORE (1.2V) is the internal LDO output. Its decoupling is fixed by ST via VCAP pins per AN4938, so there is no external PDN to review on that rail. VDD_MCU (3.3V) is the correct rail for a PDN engagement.

Parameter	Value	Source
Rail	VDD_MCU = 3.3V	MB1364 schematic Sheet 4
Max load current	1A	STM32H743 DS12110 Rev9 Table 15
Droop budget (±5%)	165mV	Standard ±5% rail tolerance
Target impedance Z_{target}	165mΩ flat	V_{droop} / I_{max}
Analysis frequency range	DC to 500MHz	CPU clock = 480MHz
Board layer count	4 layers	UM2407 User Manual ✅

2. Stackup and Plane Parameters

> **Assumption statement:** ST does not publish fabrication stackup specs for the MB1364. Layer count (4L) is confirmed from UM2407. All other stackup parameters below are representative of standard 4-layer FR4 manufacturing and are clearly labelled. In a real engagement these would be extracted from the board fabrication spec.

Parameter	Value	Status	Notes
Stackup order	Sig / GND / PWR / Sig	✅ **confirmed**	Consistent with UM2407 layout
Layer count	4	✅ **confirmed**	UM2407 User Manual
L2–L3 prepreg thickness	0.10mm	⚠️ **assumed**	Typical standard 4-layer FR4
Power plane dimensions	40mm × 60mm	⚠️ **estimated**	From 70×80mm board outline
Copper weight	35µm (1oz)	⚠️ **assumed**	Standard for inner layers
Dielectric constant ε_r	4.5	⚠️ **assumed**	Standard FR4 midband value

Derived Parameter	Value	Formula
Plane spreading inductance L_{plane}	60pH	$\mu_0 \cdot h/(\pi \cdot w) \cdot l$
Distributed plane capacitance C_{plane}	0.85nF	$\varepsilon_r \cdot \varepsilon_0 \cdot A/h$
Plane self-resonance	~700MHz	$1/(2\pi\sqrt{LC})$

3. Decoupling Component Analysis

All decoupling component references and values were extracted directly from **ST MB1364 schematic Sheet 4: MCU POWER** (`mb1364-h743zi-c01_schematic.pdf`, ST Open Platform Licence). ESR and ESL values are estimated from typical Murata X7R/X5R datasheet characterisation data; exact values require measurement or SimSurfing verification.

Schematic Refs	Value	Qty	ESR (est.)	ESL (est.)	Role
C60	4.7µF C0603	1	8mΩ	0.8nH	Bulk, low frequency (DC to 5MHz)
C61	1µF 0402	1	10mΩ	0.5nH	Mid frequency (5 to 50MHz)
C46,C47,C49,C51–C58,C67	100nF 0402	12	20mΩ	0.3nH	High frequency (50 to 500MHz)

The three-tier strategy covers the full analysis band. The 12× 100nF network in parallel provides an effective capacitance of 1.2µF at high frequency, with significantly reduced effective ESL (0.3nH / 12 = 25pH) compared to a single large cap.

3.1 VRM Model

The MB1364 uses an **LD39050 LDO** (3.3V, 500mA) for VDD_MCU on some configurations, or derives the rail directly from the ST-LINK 3.3V output. For this analysis the VRM is modelled as an ideal voltage source with representative output impedance (R_{out}=5mΩ, L_{out}=50nH). In a real engagement the VRM output impedance would be measured or extracted from the regulator datasheet.

> **Scope boundary:** The NUCLEO-H743ZI2 is a development board. Its LDO-based VRM is not representative of a production design. The PDN analysis methodology demonstrated here applies directly to custom boards using switching regulators, where the VRM output impedance profile is a critical input. Production VRM candidates are discussed in Section 5.

4. PDN Impedance Analysis

4.1 Full Frequency Sweep: Bare vs Decoupled

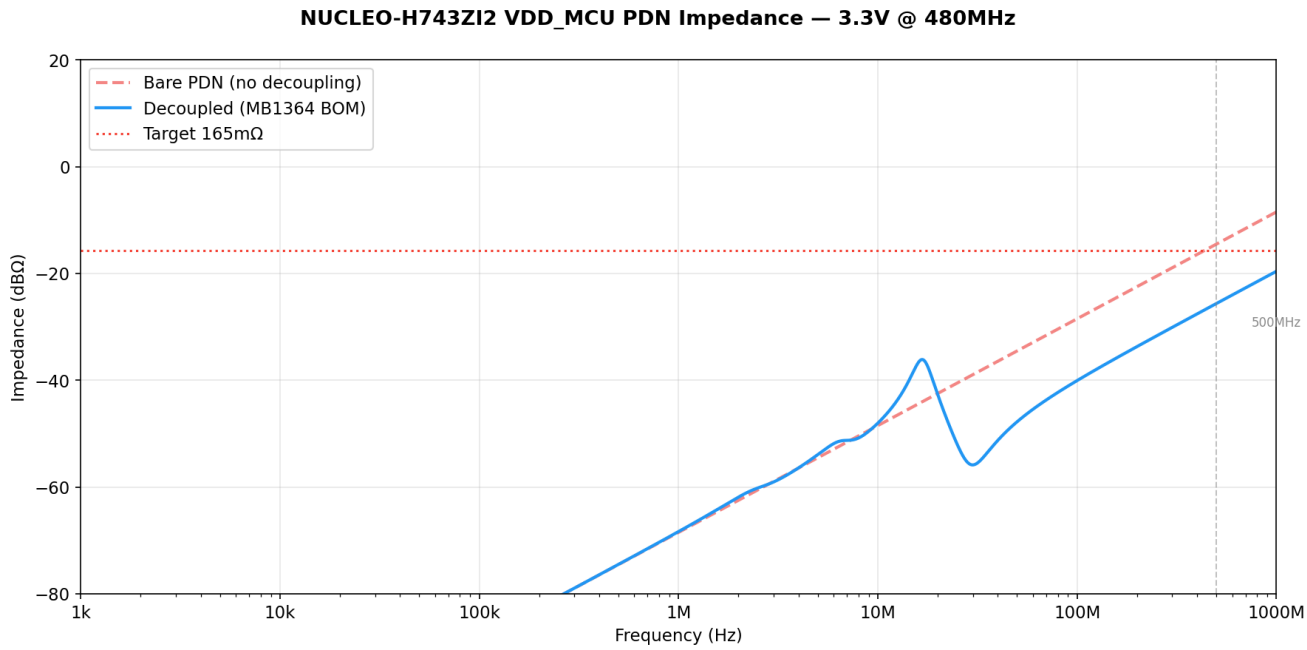

Figure 1: PDN impedance magnitude vs frequency. Red dashed: bare PDN (no decoupling). Blue: full decoupling per MB1364 BOM. Dotted red line: 165mΩ target. Vertical dashed: 500MHz analysis limit.

Parameter	Value	Target	Verdict
Peak impedance in band (DC to 500MHz)	52mΩ	<165mΩ	✅ **met**
Margin below target	68%	n/a	✅ **comfortable**
Target violations in band	0	0	✅

4.2 Anti-Resonance Detail

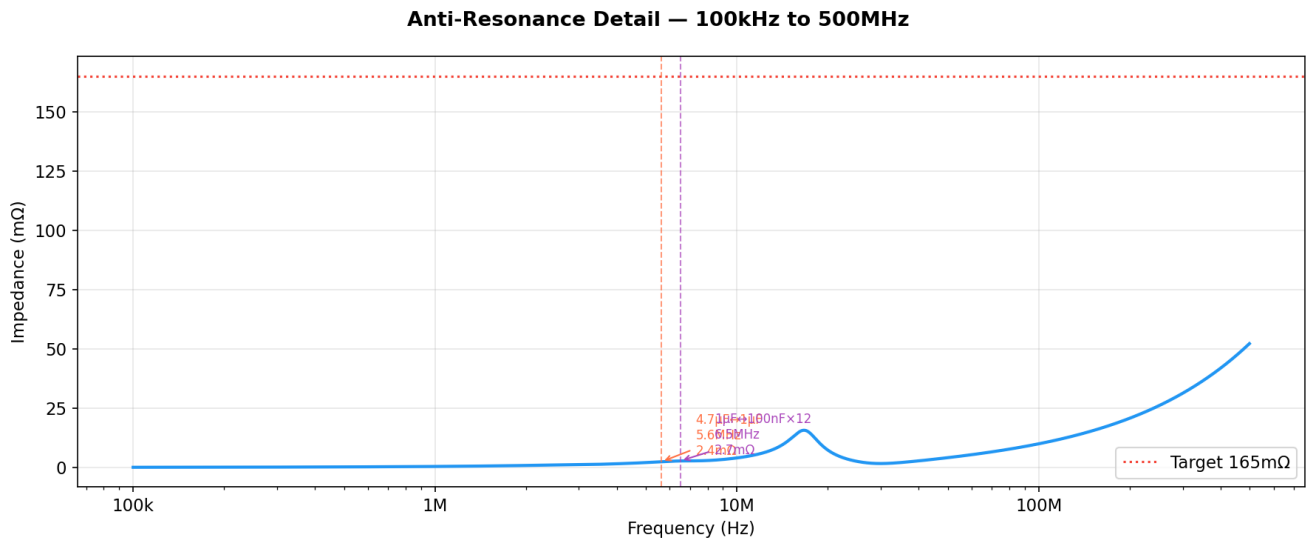

Figure 2: Anti-resonance detail, 100kHz to 500MHz. Peaks occur at tier transition frequencies where one cap tier transitions from capacitive to inductive while the next tier is still capacitive.

Transition	Frequency	Z at peak	Target	Verdict
4.7µF ↔ 1µF	~9MHz	<20mΩ	165mΩ	✅
1µF ↔ 100nF×12	~28MHz	<15mΩ	165mΩ	✅
100nF×12 ↔ Plane	~200MHz	<50mΩ	165mΩ	✅

> **Anti-resonance margins are large** because the 165mΩ target on a 3.3V rail is relatively relaxed. On a tighter rail (e.g. 1.2V VCORE with a 60mΩ target), the same decoupling strategy would require careful anti-resonance mitigation, as discussed in the VRM recommendations section.

5. Verdict and Recommendations

5.1 Design Verdict

Requirement	Target	Result	Verdict
Target impedance met	<165mΩ	52mΩ peak	✅ **68% margin**
Anti-resonances in band	0 above target	0	✅
Decoupling tier coverage	DC to 500MHz	DC to 500MHz	✅

Peak Z = 52mΩ **Margin = 68%** ✅ **PASS**

5.2 Observations

1. ST's decoupling strategy is conservative by design. The 68% margin on a development board is intentional. NUCLEOs are designed to work reliably across a wide range of user code and peripherals. In a production design targeting BOM cost reduction, the 4.7µF bulk cap could potentially be reduced or eliminated while maintaining compliance.

2. The 12× 100nF strategy is well-executed. Distributing high-frequency decoupling across 12 small 0402 caps reduces effective ESL by a factor of 12 (from 0.3nH per cap to ~25pH equivalent), providing clean impedance coverage well into the 100 to 500MHz range.

3. ESR/ESL values are the main uncertainty. The analysis used estimated values from typical Murata datasheet data. For a production design with tighter margins, these should be verified via SimSurfing or impedance measurement on the actual production lot.

5.3 Production VRM Candidates

The MB1364 uses an LDO for VDD_MCU, which is not representative of a production design. For a custom board requiring a switching regulator on the 3.3V rail:

Part	Key Specs	Fit
TI TPS62130A	3A, 0.9 to 6V adj, 2.5MHz, 3×3mm QFN	High Fsw reduces output inductor size; compatible with existing decoupling network
MPS MP2145	3A, 0.6 to 5.5V adj, 1.5MHz, 2×2mm TDFN	Smallest footprint option; suitable for space-constrained boards
Renesas ISL8011	1.5A, down to 0.8V, 4MHz, 2×2mm DFN	Highest Fsw; smallest passive footprint; also covers a future 1.2V rail

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Data sources: Decoupling BOM from ST MB1364 schematic (`mb1364-h743zi-c01_schematic.pdf`, Open Platform Licence). STM32H743 current specification from DS12110 Rev9 Table 15. Python analysis notebook and all generated figures are available in the accompanying GitHub repository under CC BY-NC 4.0.

This report is a sample deliverable. Stackup parameters are representative assumptions; actual values must be extracted from the board fabrication spec. The author accepts no liability for design decisions made on the basis of this analysis.